

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer (Node.js/TypeScript, 6+ years) who builds agentic and LLM-powered systems with production reliability discipline. Author of [career-ops](#), an open-source agentic pipeline built on Claude Code — specialized subagents, tool orchestration, structured-output contracts, ground-truth verification rules, and deterministic integrity scripts where correctness matters. Shipped a production LLM integration (OpenAI + Google Vision prescription OCR over the WhatsApp Business API) and led a zero-downtime migration of 20+ production applications reporting directly to the CTO. I decide deliberately when a problem needs an LLM, deterministic software, or a human gate — and I use coding agents as my daily engineering workflow, not an occasional assistant.

CORE COMPETENCIES

Agentic Systems & LLM Orchestration

Tool Calling & Structured Outputs

TypeScript & Node.js

AI-Assisted Development (Claude Code)

Production Reliability & Failure Handling

Workflow Automation

SQL & PostgreSQL-family Databases

Evaluation & Verification Design

WORK EXPERIENCE

MTailor Sep 2022 – Present

Software Engineer — US Remote

- Led the end-to-end migration of **20+ applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO
- Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub for message processing
- Integrated software components and third-party services across the stack; migrated a Webflow website from a Parse backend to Firebase
- Implemented Express Shipping, generating an additional \$40 revenue per order; built a 3D visualization feature (video overlay + ffmpeg) that increased buyer conversion
- Saved the company \$5,000/month by migrating all services off AWS; moved file storage from Amazon S3 to GCS via a Python script

MTailor (Contract) Mar 2022 – Jul 2022

Backend Migration Contractor — US Remote

- Built a migration script reading from MongoDB and writing to Firestore after complex processing logic
- Wrote technical documentation for fellow engineers on the new Firebase SDKs (Node.js and Python); trained the Ops team on dashboards and backup procedures

Dr Wealth Aug 2021 – Jan 2022

Software Engineer — Singapore Remote

- Led optimization of a PWA delivering real-time stock market prices; built jobs querying **2M+ Firestore records** to keep prices current from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js; added responsive React UI pages with Tailwind

Mind2matter Jan 2022 – Jul 2022

Software Engineer (Full-Stack) — US Remote

- Built backends for DeFi applications using Web3 and Node.js; delivered React UIs from Figma designs under tight timelines for demanding clients

CodeBits Jan 2020 – Jul 2021

Software Engineer / Team Lead — Uganda

- Led a team of 4 developers building legal tech systems for NGOs in Uganda; built case management for FIDA Uganda and a paralegal database for LASPNET
- Architected a microservices backend using **Apache Kafka, Docker, and Kubernetes** for FIDA Uganda's case management app
- Built a USSD service for legal aid providers communicating with a Node.js backend via gRPC — production software for African-market constraints
- Developed a cross-platform mobile app with React Native and Expo (iOS & Android) with Firebase push notifications

PROJECTS

career-ops — Open-Source Agentic Pipeline

Multi-step agentic system built on Claude Code: specialized subagents with structured-output (TSV) contracts, browser-based ground-truth verification rules, batch workers with fallback modes, and deterministic merge/dedup integrity scripts. LLM where judgment matters, code where correctness matters, human gate before every irreversible action.

Claude Code, Node.js, Playwright, agent orchestration, structured outputs

Healthcare WhatsApp Chatbot (Production LLM System)

Integrated OpenAI and Google Vision for prescription OCR via the WhatsApp Business API — probabilistic pipeline with real users, real failure modes, and vendor-API edge-case handling.

Node.js, WhatsApp Business API, OpenAI, Google Vision

Real-time MongoDB ↔ Firestore Sync Pipeline

Live two-way sync supporting parallel production traffic during a 20+ app migration — idempotent message processing over Google Pub/Sub, zero downtime.

Node.js, MongoDB, Firestore, Google Pub/Sub

EDUCATION

BSc Computer Engineering — Makerere University

2018 – 2022

Kampala, Uganda

SKILLS

Proficient: Node.js, TypeScript/JavaScript, React, MongoDB, SQL (MySQL), Express.js, Firebase, GCP, Python, Git, Linux

Intermediate: Go, AWS, Rust, React Native **Infrastructure:** Docker, Kubernetes, Apache Kafka, NATS Streaming, gRPC

Languages: English, Kiswahili